Hall Ticket Number:

MR-21

Code: B413CB5

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Regular Examinations, Aug/Sep-2023

Software Engineering

(Computer Science and Business System)

Time: 3 Hours | *Answer ALL Questions* | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Differentiate the characteristics of Software and Hardware.	2	4	1	1
1. b)	What is a prototype?	2	1	1	2
1. c)	What are non-functional requirements for a given software?	2	1	2	2
1. d)	What is a model for a given software?	2	1	2	1
1. e)	Distinguish between cohesion and coupling.	2	3	3	2
1. f)	Define a component in a software.	2	1	3	2
1. g)	How testing is different from debugging?	2	4	4	1
1. h)	Define quality for a given software.	2	1	4	2
1. i)	How software is measured?	2	1	5	1
1. j)	What is risk refinement?	2	2	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Define software myth. Explain different types of software myths.	4	1	1	1
2. b)	Explain the activities performed in unified process model.	6	4	1	1
	OR				
2. c)	Elaborate on software process framework.	4	2	1	2
2. d)	Explain spiral process model with a neat sketch.	6	1	1	2
3. a)	Elaborate on Requirements elicitation and analysis.	6	2	2	2
3. b)	What is the importance of data modelling of a system? Explain.	4	4	2	1
	OR				
3. c)	Explain the structure of a Software Requirements Document.	4	3	2	1
3. d)	Explain how the behaviour of a system can be modelled with an example.	6	5	2	1
4. a)	List out different architectural styles for a given system.	5	2	3	1
4. b)	Explain different diagrams in the conceptual model of UML.	5	4	3	1
	OR				
4. c)	Explain how system models can be translated to design models.	4	4	2	1
4. d)	Develop a class diagram for online course registration system.	6	6	3	1
5. a)	Explain the process of debugging.	5	1	4	1
5. b)	Analyze the metrics for design model and source code?	5	4	4	1
	OR				
5. c)	What is black box testing? Explain the methods.	6	2	4	1
5. d)	Explain different metrics for testing and maintenance.	4	1	4	1
6. a)	What is RMMM plan? Explain.	5	2	5	1
6. b)	Explain statistical quality assurance.	5	1	5	2
	OR				
6. c)	Explain the metrics for software measurement and quality.	4	1	5	2
6. d)	Explain how a formal technical review meeting is conducted.	6	2	5	1

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome